

LINUX AUTHENTICATION AND SECURITY LOG ANALYSIS

Cybersecurity Log Analysis Project

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Assessment Environment: Kali Linux

Tools Used: Systemd Journal (journalctl), OpenSSH, Linux authentication logs

Project Type: SOC

Educational Cybersecurity Laboratory Project

Executive Summary

This project presents a controlled analysis of Linux authentication and security logs collected from a Kali Linux laboratory environment. The objective was to identify and interpret successful and failed authentication events, user sessions, and indicators of potentially suspicious activity. The investigation used the systemd journal and OpenSSH logs to examine authentication activity involving SSH, sudo, su, and the graphical login environment. Multiple successful SSH logins and repeated failed authentication attempts were identified. The repeated SSH failures demonstrated how authentication logs can be used to identify patterns that may indicate password-guessing or brute-force activity in a real-world environment. Because the activity was intentionally generated from the local loopback address during a controlled laboratory exercise, it was treated as simulated security activity rather than an external attack. The project demonstrates practical skills in Linux log analysis, authentication monitoring, event correlation, security assessment, and SOC-oriented investigation.

LABORATORY ENVIRONMENT

The assessment was conducted in a controlled virtual laboratory environment using Kali Linux. The system was used as both the security analysis workstation and the source of the authentication and security log data examined during the investigation.

Component	Description
Operating System	Kali Linux
Authentication/Logging	Systemd journal (journalctl)
Remote Access Service	OpenSSH
SSH Port	TCP 22
Analysis Environment	Virtualized laboratory
Primary Log Source	System journal
Authentication Mechanisms Reviewed	SSH, sudo, su, and graphical authentication

The investigation focused on authentication-related events recorded by the system journal. Because the Kali installation did not contain the traditional `/var/log/auth.log` file, `journalctl` was used to retrieve and analyze the relevant authentication events.

The laboratory activity included controlled successful and failed SSH authentication attempts against the local Kali system. These events were intentionally generated to demonstrate authentication-log investigation techniques.

METHODOLOGY

The assessment followed a controlled log-analysis methodology designed to identify authentication activity and demonstrate how security analysts can investigate Linux authentication events.

1. System and Service Preparation

The Kali Linux environment was prepared for analysis, and the OpenSSH service was enabled and started. The SSH service was confirmed to be active and listening on TCP port 22.

2. Log Source Identification

The traditional `"/var/log/auth.log"` file was checked but was not present on the system. The systemd journal was therefore selected as the primary source for authentication and security events.

3. Successful Authentication Testing

A controlled SSH connection was established from the Kali system to its own localhost address. A successful password authentication was generated and recorded in the system journal.

4. Failed Authentication Testing

Controlled failed SSH authentication attempts were generated using an incorrect password. The resulting authentication failures were recorded and reviewed to demonstrate how repeated failed login attempts appear in security logs.

5. Session Analysis

The `"who"` and `"last"` commands were used to examine active and recent user sessions. This provided additional context for correlating authentication events with user activity.

6. Broader Authentication Analysis

The system journal was searched for authentication-related events involving SSH, sudo, su, and the graphical authentication environment. This allowed authentication failures from different mechanisms to be compared.

7. Security Interpretation

The collected events were reviewed for patterns that could indicate suspicious authentication activity. Repeated SSH authentication failures were identified as a potential brute-force indicator in a real-world environment. However, the SSH failures in this assessment were intentionally generated from the local loopback address (`"::1"`) as part of the controlled laboratory exercise.

8. Documentation

The findings were documented together with their security implications, potential impact, risk level, and recommended security controls.

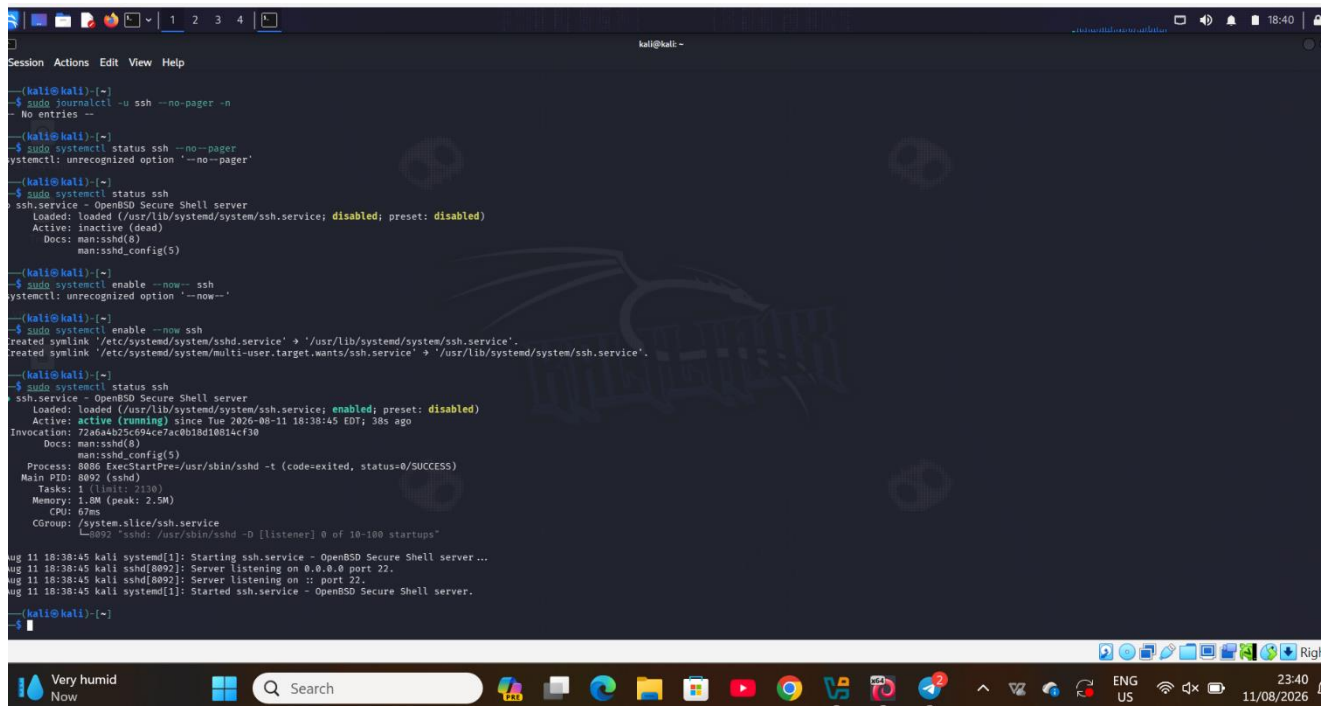
3. LOG ANALYSIS FINDINGS

3.1 SSH Service Status

The OpenSSH service was enabled and started during the assessment. The service status confirmed that SSH was active and running on the Kali Linux system.

The SSH daemon was listening on TCP port 22 on both IPv4 ("0.0.0.0") and IPv6 ("::"). This confirmed that the system was accepting SSH connections.

The service status was recorded as evidence before authentication testing was performed.



```
kali@kali:~$ sudo journalctl -u ssh --no-pager -n
No entries --

kali@kali:~$ sudo systemctl status ssh --no-pager
systemctl: unrecognized option '--no-pager'

kali@kali:~$ sudo systemctl status ssh
ssh.service - OpenSSH Secure Shell server
Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: disabled)
Active: inactive (dead)
Docs: man:sshd(8)
      man:sshd_config(5)

kali@kali:~$ sudo systemctl enable --now ssh
systemctl: unrecognized option '--now--'

kali@kali:~$ sudo systemctl enable --now ssh
Created symlink '/etc/systemd/system/ssh.service' → '/usr/lib/systemd/system/ssh.service'.
Created symlink '/etc/systemd/system/multi-user.target.wants/ssh.service' → '/usr/lib/systemd/system/ssh.service'.

kali@kali:~$ sudo systemctl status ssh
ssh.service - OpenSSH Secure Shell server
Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: disabled)
Active: active (running) since Tue 2026-08-11 18:38:45 EDT; 38s ago
Invocations: 726d4425c59ace7ac6b18d10814cf30
Docs: man:sshd(8)
      man:sshd_config(5)
Process: 8086 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
Main PID: 8092 (sshd)
Tasks: 1 (limit: 2130)
Memory: 1.0M (peak: 2.5M)
CPU: 67ms
CGroup: /system.slice/ssh.service
        └─8092 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

aug 11 18:38:45 kali systemd[1]: Starting ssh.service - OpenSSH Secure Shell server...
aug 11 18:38:45 kali sshd[8092]: Server listening on 0.0.0.0 port 22.
aug 11 18:38:45 kali sshd[8092]: Server listening on :: port 22.
aug 11 18:38:45 kali systemd[1]: Started ssh.service - OpenSSH Secure Shell server.

kali@kali:~$
```

OpenSSH service active and listening on TCP port 22

3.2 Successful SSH Authentication

A controlled SSH connection was established from the Kali Linux system to its local loopback address ("::1"). The authentication log recorded successful password authentication for the "kali" user.

The event was recorded at 18:46:58 on August 11, followed by a session-opening event for the authenticated user. A second successful SSH authentication was also recorded at 18:53:42. The source address ("::1") represents the local loopback interface. Therefore, these successful authentication events were generated locally as part of the controlled laboratory exercise and do not represent unauthorized external access.

```
(kali@kali)-[~]
$ ssh kali@localhost
The authenticity of host 'localhost (::1)' can't be established.
ED25519 Key fingerprint is: SHA256:17CqA245G19K3m04ffdyvQmYscU1bVwcZqg+AuS4
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'localhost' (ED25519) to the list of known hosts.
kali@localhost's password:
Linux kali 7.0.12-kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 7.0.12-2kali1 (2026-06-18) x86_64

The programs included with the Kali GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Kali GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
(kali@kali)-[~]
$ exit
Connection to localhost closed.

(kali@kali)-[~]
$ sudo journalctl -u ssh --no-pager -n 30
Aug 11 18:38:45 kali systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Aug 11 18:38:45 kali sshd[8092]: Server listening on 0.0.0.0 port 22.
Aug 11 18:38:45 kali sshd[8092]: Server listening on :: port 22.
Aug 11 18:38:45 kali systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
Aug 11 18:46:50 kali sshd-session[11482]: Accepted password for kali from ::1 port 55290 ssh2
Aug 11 18:46:50 kali sshd-session[11482]: pam_unix(sshd:session): session opened for user kali(uid=0) by kali(uid=0)

(kali@kali)-[~]
$
```

Successful SSH authentication recorded in the system journal

3.3 Repeated Failed SSH Authentication

Multiple failed SSH authentication attempts were recorded for the "kali" account during the controlled test.

At 18:55:53, the system recorded an authentication failure. This was followed by failed password attempts at 18:55:55, 18:56:23, and 18:57:37. The SSH connection was subsequently closed at 18:57:38, and the PAM subsystem recorded additional authentication failures.

The repeated failures occurred within a short period and demonstrate an authentication-failure pattern that could be an indicator of password-guessing or brute-force activity in a production environment.

However, the source address for these events was "::1", the local loopback address. The attempts were intentionally generated during this controlled laboratory exercise and therefore should not be classified as evidence of an actual external attack.

Key evidence:

- "authentication failure"
- "Failed password for kali from ::1"
- "PAM 2 more authentication failures"
- "Connection closed by authenticating user kali"

Security significance: Repeated authentication failures should be monitored and investigated in production environments, particularly when they originate from external IP addresses or target multiple accounts.

```
Aug 11 18:40:58 kali sshd-session[11482]: Accepted password for kali from ::1 port 55258 ssh2
Aug 11 18:46:58 kali sshd-session[11482]: pam_unix(sshd:session): session opened for user kali(uid=1000) by kali(uid=0)
Aug 11 18:53:42 kali sshd-session[15476]: Accepted password for kali from ::1 port 54916 ssh2
Aug 11 18:53:42 kali sshd-session[15476]: pam_unix(sshd:session): session opened for user kali(uid=1000) by kali(uid=0)

--(kali@kali)-[~]
└─$ ssh kali@localhost
kali@localhost's password:
Permission denied, please try again.
kali@localhost's password:
Permission denied, please try again.
kali@localhost's password:
Permission denied (publickey,password).

--(kali@kali)-[~]
└─$ sudo journalctl -u ssh --no-pager -n 30

Aug 11 18:38:45 kali systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Aug 11 18:38:45 kali sshd[8092]: Server listening on 0.0.0.0 port 22.
Aug 11 18:38:45 kali sshd[8092]: Server listening on :: port 22.
Aug 11 18:38:45 kali systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
Aug 11 18:46:58 kali sshd-session[11482]: Accepted password for kali from ::1 port 55258 ssh2
Aug 11 18:46:58 kali sshd-session[11482]: pam_unix(sshd:session): session opened for user kali(uid=1000) by kali(uid=0)
Aug 11 18:53:42 kali sshd-session[15476]: Accepted password for kali from ::1 port 54916 ssh2
Aug 11 18:53:42 kali sshd-session[15476]: pam_unix(sshd:session): session opened for user kali(uid=1000) by kali(uid=0)
Aug 11 18:55:53 kali unix_chkpwd[16687]: password check failed for user (kali)
Aug 11 18:55:53 kali sshd-session[16613]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=:1 user=kali
Aug 11 18:55:55 kali sshd-session[16613]: Failed password for kali from ::1 port 47962 ssh2
Aug 11 18:56:20 kali unix_chkpwd[16904]: password check failed for user (kali)
Aug 11 18:56:23 kali sshd-session[16613]: Failed password for kali from ::1 port 47962 ssh2
Aug 11 18:57:35 kali unix_chkpwd[17514]: password check failed for user (kali)
Aug 11 18:57:37 kali sshd-session[16613]: Failed password for kali from ::1 port 47962 ssh2
Aug 11 18:57:38 kali sshd-session[16613]: Connection closed by authenticating user kali ::1 port 47962 [preauth]
Aug 11 18:57:38 kali sshd-session[16613]: PAM 2 more authentication failures; logname= uid=0 euid=0 tty=ssh ruser= rhost=:1 user=kali

--(kali@kali)-[~]
└─$
```

Repeated failed SSH authentication events recorded in the system journal

3.4 Failed sudo Authentication

The system journal recorded a failed authentication attempt involving the "sudo" privilege-escalation mechanism on August 3.

The event was recorded as:

"pam_unix(sudo:auth): authentication failure"

The event involved the "kali" user attempting to authenticate through "sudo". A failed "sudo" authentication does not by itself demonstrate malicious activity, but repeated or unexpected failures could indicate an attempt to obtain elevated privileges or unauthorized use of an account.

In this controlled assessment, the event was treated as an authentication-security event rather than evidence of an actual privilege-escalation attack.

Security significance: Failed privileged authentication attempts should be monitored because successful unauthorized authentication through "sudo" could allow a user to execute commands with elevated privileges.

3.5 Failed su Authentication

The system journal also recorded a failed authentication attempt involving the "su" command on August 9.

The event was recorded as:

"pam_unix(su:auth): authentication failure"

The event involved the "kali" user attempting to authenticate as the "root" account. Because the root account has extensive privileges, unexpected attempts to switch to root should be reviewed carefully in a production environment.

The recorded event does not establish that unauthorized privilege escalation occurred. It demonstrates that an authentication failure was detected while attempting to switch user context.

Security significance: Repeated failed attempts to authenticate as root could indicate attempts to obtain privileged access and should therefore be monitored and investigated.

3.6 Other Authentication Failures

The broader system journal search identified additional authentication failures associated with the XFCE screensaver. Events were recorded on July 5 and July 23 and were associated with the "kali" user.

These events were recorded as:

"pam_unix(xfce4-screensaver:auth): authentication failure"

The events indicate unsuccessful authentication attempts during graphical screen-unlock activity. However, the available log information does not establish whether these were caused by incorrect passwords, user error, or malicious activity.

The events were therefore treated as authentication anomalies rather than confirmed security incidents.

Security significance: Repeated authentication failures across different authentication mechanisms should be monitored and correlated with other security events to determine whether they represent normal user activity or potentially suspicious behaviour.

3.7 Authentication Event Correlation

The investigation identified authentication-related events across multiple mechanisms, including SSH, "sudo", "su", and the XFCE screensaver.

The most significant pattern was the repeated SSH authentication failures recorded between 18:55 and 18:57 on August 11. The events included multiple failed password attempts and additional PAM authentication failures.

Correlation with the session history showed that successful SSH sessions had previously been established from the local loopback address ("::1"). The repeated failed SSH attempts also originated from "::1".

Because the authentication activity was intentionally generated during the laboratory

exercise, the events are classified as simulated security-test activity. Nevertheless, the pattern demonstrates how a security analyst could identify repeated authentication failures and investigate them as a potential brute-force indicator in a real environment.

This correlation demonstrates the importance of examining authentication events together with timestamps, source addresses, usernames, authentication mechanisms, and session history rather than analyzing individual log entries in isolation.

4. RISK ASSESSMENT

The identified authentication events were assessed according to their security significance, potential impact, likelihood of indicating suspicious activity, and recommended remediation priority. Because the findings are authentication events rather than software vulnerabilities, CVSS scores are not applicable and are therefore recorded as N/A.

Finding/Event	Severity	CVSS	Potential Impact	Risk Level	Recommended Priority
Repeated SSH Authentication Failures	Medium	N/A	May indicate password-guessing or brute-force activity. Successful exploitation could result in unauthorized account access	Medium	High
Failed <i>sudo</i> Authentication	Medium	N/A	Repeated failures could indicate an attempt to obtain elevated privileges. Successful unauthorized authentication could allow privileged command execution	Medium	Medium
Failed <i>su</i> Authentication	Medium	N/A	Unauthorized successful authentication to the root account could provide extensive control over the system	Medium	High
Other Authentication Failures – XFCE Screensaver	Low	N/A	May indicate incorrect credentials or usual authentication activity. Repeated occurrence could warrant further investigation	Low	Low
Authentication Events Across Multiple Mechanisms	Medium	N/A	Authentication failures across SSH, sudo, su, and graphical authentication may indicate account misuse or credential-related issues when occurring repeatedly	Medium	High

5. RECOMMENDED SECURITY CONTROLS

The following security controls are recommended to reduce the risk identified during the Linux authentication and security log analysis

5.1 SSH Authentication Monitoring

Monitor SSH authentication logs for repeated failed login attempts, particularly when multiple failures occur within a short period.

Recommended controls:

- Monitor failed SSH authentication events.
- Configure alerts for repeated failures.
- Investigate unusual source addresses.
- Use centralized log monitoring where available
- Consider rate-limiting or temporary blocking of repeated authentication attempts.

5.2 Strong Authentication Controls

Strong authentication reduces the likelihood of unauthorized access resulting from compromised or weak credentials.

Recommended controls:

- Use strong and unique passwords.
- Disable unnecessary user accounts.
- Apply appropriate password policies.
- Use SSH keys where appropriate.
- Implement multi-factor authentication for sensitive systems where supported.

5.3 Privileged Access Management

The failed *sudo* and *su* authentication events demonstrate the importance of protecting privileged access

Recommended controls:

- Apply the principle of least privilege.
- Restrict unnecessary *sudo* privilege
- Monitor privilege authentication attempts.
- Avoid unnecessary direct root access
- Review privilege-account activity regularly

5.4 Secure Remote Access

SSH should be securely configured because it provides remote access to the Linux system.

Recommended controls:

- Disable unnecessary remote-access services.
- Restrict SSH access to authorized users.
- Use SSH keys where appropriate.
- Disable unnecessary authentication methods.
- Apply firewall rules to restrict access to TCP port 22.
- Keep the SSH server and operating system patched.

5.5 Authentication Event Monitoring and Correlation

Authentication events from different mechanisms should be reviewed together to identify patterns that may not be obvious from individual log entries.

Recommended controls:

- Monitor SSH, *sudo*, *su*, and graphical authentication events.
- Maintain accurate system timestamps.
- Review authentication logs regularly.
- Correlate usernames, timestamps, source addresses, and session activity.
- Use a SIEM or centralized logging solution in larger environments.

5.6 Incident Response for Suspicious Authentication Activity

Repeated authentication failures should trigger an appropriate investigation process when observed in a production environment.

Recommended controls:

- Verify whether the source address is authorized.
- Determine which account was targeted.
- Review successful and failed authentication events.
- Check for suspicious activity following a successful login.
- Temporarily restrict affected accounts when compromise is suspected.
- Document and escalate confirmed security incidents.

5.7 Log Retention and Protection

Authentication logs should be retained and protected from unauthorized modification.

Recommended controls:

- Establish appropriate log-retention periods.
- Restrict access to security logs.
- Forward important logs to a centralized logging platform.
- Protect logs against unauthorized modification or deletion.
- Regularly review logging configuration.

6. RECOMMENDED SECURITY POLICIES

The following policies are recommended to establish consistent security practices for authentication, privileged access, remote access, and security-log monitoring.

6.1 Password and Authentication Policy

The organization should establish requirements for strong authentication and secure credential management.

The policy should require:

- Strong and unique passwords.
- Protection of user credentials.
- Regular review of user accounts.
- Immediate disabling of unnecessary or inactive accounts.
- Multi-factor authentication for systems that support it.
- Monitoring and investigation of repeated authentication failures.

6.2 Secure Remote Access Policy

Remote access to Linux systems should be restricted to authorized users and approved services.

The policy should require:

- SSH as the preferred secure remote-access protocol.
- Prohibition of insecure remote-access services where possible.
- Restriction of SSH access to authorized users.
- Appropriate firewall controls for remote-access ports.
- Regular patching of remote-access software.
- Monitoring of remote authentication activity.

6.3 Privileged Access Management Policy

Privileged access should be limited to authorized personnel and used only when necessary.

The policy should require:

- Least-privilege access.
- Controlled use of *sudo* privileges.
- Regular review of privileged accounts.
- Monitoring of privileged authentication attempts.
- Prompt removal of unnecessary administrative privileges.

6.4 Security Logging and Monitoring Policy

Security-relevant authentication events should be logged, protected, retained, and reviewed.

The policy should require:

- Logging of successful and failed authentication attempts.
- Monitoring of SSH, *sudo*, *su*, and other important authentication mechanisms.
- Protection of logs-retention periods.

- Appropriate logs-retention periods.
- Regular review of authentication events.
- Centralized logging or SIEM integration where appropriate.

6.5 Incident Response Policy

The organization should establish procedures for responding to suspicious authentication activity.

The policy should define procedures for:

- Identifying repeated authentication failures.
- Investigating suspicious source addresses.
- Determining whether an account has been compromised
- Containing suspected unauthorized access.
- Escalating confirmed security incidents.
- Documenting investigation and remediation activities.

6.6 Vulnerability and Patch Management Policy

Operating system and security-related services should be maintained using a structured vulnerability and patch-management process.

The policy should require:

- Regular vulnerability assessment.
- Timely installation of security updates.
- Tracking of identified vulnerabilities.
- Prioritization of remediation according to risk.
- Verification that remediation has been completed.
- Documentation of exceptions where vulnerabilities cannot immediately be remediated.

7. CONCLUSION

This project demonstrated the practical process of analysing Linux authentication and security logs within a controlled Kali Linux laboratory environment. The investigation used the `systemd journal` and `OpenSSH` logs to identify successful authentication events, repeated failed SSH attempts, privileged authentication failures involving `"sudo"` and `"su"`, and additional authentication events from the graphical environment.

The analysis demonstrated how timestamps, usernames, source addresses, authentication mechanisms, and session history can be correlated to identify patterns that may require further investigation. The repeated SSH authentication failures provided a practical example of how potential password-guessing or brute-force indicators can be identified from security logs.

Because the authentication activity was intentionally generated within the laboratory environment and originated from the local loopback address, the events were treated as simulated security activity rather than evidence of an actual external attack.

Overall, the assessment demonstrates practical skills in Linux security monitoring, authentication-log analysis, event correlation, risk assessment, security controls, and incident-response thinking. These techniques can be applied in a Security Operations Centre (SOC) environment to support the detection and investigation of suspicious authentication activity.